

GUNGUN AGGARWAL

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EDUCATION

Lovely Professional University

B.Tech - Computer Science and Engineering - **CGPA - 7.07**

2022 – 2026

Phagwara, Punjab

COURSEWORK / SKILLS

- | | | | |
|-------------------------------------|-------------------------------------|---------------------------------|---------------------------------|
| • Data Structures & Algorithms | • Database Management System (DBMS) | • REST APIs & Microservices | • Docker & Containerization |
| • Object Oriented Programming (OOP) | • Computer Networks | • Spring Boot & Spring Security | • Redis & RabbitMQ |
| • Operating System | • Software Engineering | • Machine Learning | • Git, GitHub & Version Control |
| | • Machine Learning | | |

EXPERIENCE

GeeksforGeeks- Data Structures & Algorithms- Self Paced 

June 2024 - July 2024

Role - *DSA Trainee*

Phagwara, Punjab

- Gained hands-on experience in core programming concepts and problem-solving techniques.
- Implemented key data structures such as arrays, linked lists, stacks, queues, trees and graphs which enhanced my algorithmic thinking for optimized code development.
- Engineered an efficient Sudoku Solver using the backtracking algorithm to recursively fill a 9x9 grid while enforcing all Sudoku rules, showcasing mastery in recursion, constraint satisfaction, and problem-solving techniques in data structures and algorithms.

PROJECTS

Smartsure Insurance Management System | Java, Spring Boot, Angular, MySQL, Docker ,JWT

- Built a microservices-based Insurance Management System using **Spring Boot** and **Angular** with modules for policy purchase, claim processing, and admin management.
- Implemented secure authentication using **JWT** and **OTP** verification, along with API documentation using **Swagger/OpenAPI** for efficient backend testing.
- Integrated **RabbitMQ** for asynchronous communication, **Redis** for caching, and **Docker** for containerization, while using **Zipkin** for distributed request tracing.

Air Quality Monitoring and Prediction System  | Python, TensorFlow, Keras, RNN,LSTM,GRU

- Built an end-to-end Air Quality Monitoring and Prediction system using **Streamlit** and real-time **AQI APIs**, enabling live pollutant tracking and predictive analysis.
- Implemented and compared **RNN**, **LSTM**, and **GRU** deep learning models for AQI forecasting, achieving R^2 score up to 0.999, demonstrating strong time-series modeling capability.
- Designed a data preprocessing and neural sequence transformation pipeline to enable recurrent networks to effectively model temporal dependencies in AQI data.

Music Recommendation System  | Python, Machine Learning, UI elements

- Designed a Music Recommendation System using **Streamlit** with a content-based approach powered by **KNN**.
- Preprocessed a **Kaggle.com** dataset of 170k+ songs by scaling audio features.
- Developed an interactive **Streamlit** app allowing users to set preferences via sliders.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, SQL, C++

Frameworks & Technologies: Spring Boot, Spring Security, Angular, Microservices, HTML, CSS, REST APIs, JWT

Databases & Tools: MySQL, Redis, RabbitMQ, Docker, Git, GitHub, Swagger, Zipkin, VS Code, Postman

CERTIFICATIONS

- Generative AI for everyone
- Introduction To Large Language Models
- Programming in C++: A Hands-on Introduction

EXTRACURRICULAR

- **Event Manager**, Ojaswi Club LPU (jan 2024 – july 2024)